C2 Notes: exponential and logarithmic functions

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Exponential functions describe anything that grows or shrinks by the same *factor* in equal time steps: bacteria, money with interest, radioactive material, the cooling of tea. Logarithms answer the reverse question: *how long* until a given amount is reached? Together they also explain why scales like decibels and pH work the way they do.

Notation. Dutch books write the logarithm with base g as ${}^{g}\log x$ (*grondtal* g). English books write $\log_g x$. Both mean the same. On the calculator, log is ${}^{10}\log$ and ln is ${}^{e}\log$.

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1. Exponential functions

An **exponential function** has the form

$$f(x) = b \cdot g^x, \qquad g > 0,\ g \neq 1.$$

- $b = f(0)$ is the **initial value** (*beginwaarde*).
- g is the **growth factor** (*groeifactor*): each time x increases by 1, f is multiplied by g, because $f(x+1) = bg^{x+1} = g \cdot f(x)$.
- $g > 1$: exponential growth; $0 < g < 1$: exponential decay.
- Domain $\mathbb{R}$; for $b > 0$ the range is $(0, \infty)$; the x-axis is a horizontal asymptote.

Growth of $p\%$ per step means $g = 1 + \frac{p}{100}$; a decrease of $p\%$ means $g = 1 - \frac{p}{100}$.

Why exponential and not linear?

A linear function adds the same *amount* each step: $f(x+1) - f(x)$ is constant. An exponential function multiplies by the same *factor*: $\frac{f(x+1)}{f(x)}$ is constant. To decide which model fits a table, check differences and ratios.

Changing the time unit

If the growth factor per year is g, then over n years it is g^n and per month it is $g^{1/12}$, since twelve months multiplied together must give g: $(g^{1/12})^{12} = g$.

Example 1. A population grows by 4% per year. Growth factor per decade: $1.04^{10} \approx 1.480$ (48% per decade, not 40%). Growth factor per month: $1.04^{1/12} \approx 1.00327$ (about 0.327% per month, not 0.333%).

Transformations

$f(x) = 2 \cdot 3^{x-1} - 6$ is 3^x translated 1 right, stretched by 2 from the x-axis, translated 6 down. So the horizontal asymptote is $y = -6$ and the range is $(-6, \infty)$. Note that $3^{x-1} = \frac{1}{3} \cdot 3^x$: a horizontal translation of an exponential is the same as a vertical stretch. That is special to exponentials.

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2. Definition of the logarithm

$2^x = 8$ has the solution $x = 3$. But what about $2^x = 5$? The function $x \mapsto 2^x$ is increasing and takes every positive value exactly once, so it has an inverse function. That inverse is the logarithm.

Definition. For $g > 0$, $g \neq 1$ and $x > 0$:

$$^{g}\log x = y \iff g^{y} = x.$$

In words: $^{g}\log x$ is *the exponent you must put on g to get x*.

Consequences, straight from the definition:

$$g^{\,^{g}\log x} = x \ (x > 0), \qquad ^{g}\log(g^{y}) = y, \qquad ^{g}\log 1 = 0, \qquad ^{g}\log g = 1.$$

Example 2.

- $^{2}\log 32 = 5$ because $2^{5} = 32$.
- $^{3}\log \frac{1}{9} = -2$ because $3^{-2} = \frac{1}{9}$.
- $^{4}\log 8 = \frac{3}{2}$ because $4^{3/2} = (\sqrt{4})^{3} = 8$.
- $^{1/2}\log 8 = -3$ because $\left(\frac{1}{2}\right)^{-3} = 8$.

The function $f(x) = {}^{g}\log x$ has domain $(0, \infty)$ and range $\mathbb{R}$. Its graph is the reflection of $y = g^{x}$ in $y = x$, so it has the y-axis as **vertical** asymptote and passes through $(1, 0)$ and $(g, 1)$. For $g > 1$ it is increasing; for $0 < g < 1$ decreasing.

Transformed logarithmic functions and inverses

Everything from C1 about transformations and inverses applies.

Example 2b. $f(x) = 2 + {}^{3}\log(x - 1)$.

- It is $y = {}^{3}\log x$ translated by $(1, 2)$. The vertical asymptote moves with it: $x = 1$. Domain $x > 1$, range $\mathbb{R}$.
- Zero: $^{3}\log(x-1) = -2 \iff x - 1 = 3^{-2} = \frac{1}{9} \iff x = 1\frac{1}{9}$.
- Inverse: $y = 2 + {}^{3}\log(x-1) \iff {}^{3}\log(x-1) = y-2 \iff x-1 = 3^{y-2}$. So $f^{-1}(x) = 1+3^{x-2}$, with horizontal asymptote $y = 1$: the mirror image of the vertical asymptote $x = 1$, as it should be.

A logarithmic graph has no horizontal asymptote, even though it rises very slowly: $^{10}\log x$ reaches 6 only at $x = 1\,000\,000$, but it does reach every value eventually, because $^{10}\log(10^{N}) = N$. Slow growth is not the same as levelling off.

Why logarithms with base $g > 1$ are increasing. If $0 < a < b$ and we had $^{g}\log a \geq {}^{g}\log b$, then raising g to both sides (and g^{x} is increasing for $g > 1$) would give $a \geq b$, a contradiction. The same argument with the inequality reversed shows that for $0 < g < 1$ the logarithm is decreasing. This is the fact behind every inequality in §7.

> **Common mistake.** $^{g}\log x$ only exists for $x > 0$. Every time you solve a log equation, the domain conditions must be checked at the end.

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3. Rules of logarithms, with proofs

For $g > 0$, $g \neq 1$, $a > 0$, $b > 0$ and any real p:

1. **Product rule:** $^{g}\log(ab) = {}^{g}\log a + {}^{g}\log b$
2. **Quotient rule:** $^{g}\log\left(\frac{a}{b}\right) = {}^{g}\log a - {}^{g}\log b$
3. **Power rule:** $^{g}\log(a^{p}) = p \cdot {}^{g}\log a$

Each rule is a power rule in disguise. The logarithm turns multiplication into addition because exponents add when powers are multiplied.

Proof of 1. Let $u = {}^g\log a$ and $v = {}^g\log b$. By definition $g^u = a$ and $g^v = b$. Then

$$ab = g^u \cdot g^v = g^{u+v},$$

so, by the definition of the logarithm, ${}^g\log(ab) = u + v = {}^g\log a + {}^g\log b$. □

Proof of 2. With the same u, v: $\dfrac{a}{b} = \dfrac{g^u}{g^v} = g^{u-v}$, so ${}^g\log \frac{a}{b} = u - v$. □

Proof of 3. With $u = {}^g\log a$: $a^p = (g^u)^p = g^{pu}$, so ${}^g\log(a^p) = pu = p \cdot {}^g\log a$. □

There is **no** rule for ${}^g\log(a + b)$. For example ${}^{10}\log(1 + 1) \approx 0.301$, while ${}^{10}\log 1 + {}^{10}\log 1 = 0$.

Example 3. Write as a single logarithm: $2\log x + \log 5 - \frac{1}{2}\log(25x)$ (base 10, $x > 0$).

$$= \log x^2 + \log 5 - \log\sqrt{25x} = \log \frac{5x^2}{5\sqrt{x}} = \log\left(x^{3/2}\right) = \tfrac{3}{2}\log x.$$

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4. Change of base

Calculators usually only have log (base 10) and ln (base e). To compute ${}^2\log 5$:

Change of base. For $a, g, h > 0$ with $g, h \neq 1$:

$${}^g\log a = \frac{{}^h\log a}{{}^h\log g}.$$

Proof. Let $y = {}^g\log a$, so $g^y = a$. Take ${}^h\log$ of both sides and use the power rule: $y \cdot {}^h\log g = {}^h\log a$. Since $g \neq 1$, ${}^h\log g \neq 0$, so we may divide. □

So ${}^2\log 5 = \dfrac{\log 5}{\log 2} \approx 2.322$.

Two useful consequences:

- ${}^g\log a \cdot {}^a\log g = 1$ (take $h = a$ in the formula).
- ${}^a\log b \cdot {}^b\log c = {}^a\log c$: the "chain" collapses, since $\dfrac{\log b}{\log a} \cdot \dfrac{\log c}{\log b} = \dfrac{\log c}{\log a}$.

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5. Solving exponential and logarithmic equations

There are four standard types. In every case: aim for the form "same base on both sides" or "log of something = number".

Type A: make the bases equal. $2^{2x} = 8^{x-1}$. Write $8 = 2^3$: $2^{2x} = 2^{3x-3}$. Since 2^x is one-to-one, $2x = 3x - 3$, so $x = 3$.

Type B: take a logarithm. $5^x = 12$ gives $x = {}^5\log 12 = \dfrac{\log 12}{\log 5} \approx 1.544$. Exact answer: ${}^5\log 12$.

Type C: log equals number, use the definition. ${}^2\log(x+3) = 4 \Rightarrow x + 3 = 2^4 = 16 \Rightarrow x = 13$. Check: $x + 3 = 16 > 0$.

Type D: combine logs first, then use the definition. ${}^3\log x + {}^3\log(x - 8) = 2$. Domain: $x > 0$ and $x > 8$, so $x > 8$. Combine: ${}^3\log(x(x-8)) = 2$, so $x^2 - 8x = 9$, $(x-9)(x+1) = 0$, $x = 9$ or $x = -1$. Only $x = 9$ satisfies $x > 8$.

Why the domain check matters. The combined equation ${}^3\log(x(x-8)) = 2$ is defined also for $x < 0$ (then both factors are negative and the product is positive). Combining logs *enlarges* the domain, so it can add false solutions. Always write the domain condition first.

Type E: substitution. $4^x - 6 \cdot 2^x + 8 = 0$. Note $4^x = (2^x)^2$. Put $u = 2^x > 0$: $u^2 - 6u + 8 = 0$, $u = 2$ or $u = 4$, so $x = 1$ or $x = 2$.

Example 4 (log with different bases). ${}^2\log x = {}^4\log(x+6)$. Change base: ${}^4\log(x+6) = \dfrac{{}^2\log(x+6)}{{}^2\log 4} = \frac{1}{2}\,{}^2\log(x+6)$. So $2\,{}^2\log x = {}^2\log(x+6)$, ${}^2\log x^2 = {}^2\log(x+6)$, $x^2 = x+6$, $x = 3$ or $x = -2$. Domain $x > 0$: $x = 3$.

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6. The number e and the natural logarithm

Where e comes from

Put €1 in a bank at 100% interest per year. Paid once a year you get €2. Paid twice a year, 50% each half year: $1.5^2 = 2.25$. Paid monthly: $\left(1 + \frac{1}{12}\right)^{12} \approx 2.613$. Paid n times:

$$\left(1 + \frac{1}{n}\right)^n.$$

As n grows, this increases but levels off:

n	1	2	12	365	10 000	1 000 000
$\left(1+\frac{1}{n}\right)^n$	2	2.25	2.613	2.7146	2.71815	2.718280

The limit is the number

$$e = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n = 2.718281828\,...$$

It is irrational (in fact transcendental). Continuous growth at rate r per year multiplies by e^r per year.

There is a second reason e is special, which you will see in stage D: among all exponential functions g^x, the function e^x is the only one whose graph has slope exactly 1 at $x = 0$, and as a result its derivative is itself. That is why mathematicians prefer base e.

Natural logarithm

$\ln x = {}^e\log x$. All rules above hold for ln. Key facts: $\ln 1 = 0$, $\ln e = 1$, $e^{\ln x} = x$, $\ln(e^x) = x$.

Every exponential function can be written with base e: since $g = e^{\ln g}$,

$$b \cdot g^t = b \cdot e^{t \ln g} = b \cdot e^{kt} \quad \text{with } k = \ln g.$$

For decay ($g < 1$), $k < 0$. Physics and biology texts almost always use this form.

Example 5. Solve exactly: $e^{2x} - 3e^x = 4$. Put $u = e^x > 0$: $u^2 - 3u - 4 = 0$, $(u-4)(u+1) = 0$, $u = 4$ (since $u = -1 < 0$ is impossible). So $e^x = 4$, $x = \ln 4 = 2\ln 2$.

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7. Inequalities

The key fact: if $g > 1$, then g^x and ${}^g\log x$ are **increasing**, so they preserve "$<$". If $0 < g < 1$ they are **decreasing**, so they reverse it.

Example 6. $0.8^x < 0.3$. Take log (base 10, increasing, so the sign stays): $x \log 0.8 < \log 0.3$. Now $\log 0.8 < 0$, so dividing by it **reverses** the sign: $x > \dfrac{\log 0.3}{\log 0.8} \approx 5.40$.

Safer method that always works: solve the equation $0.8^x = 0.3$ exactly ($x = {}^{0.8}\log 0.3$), then look at the graph: 0.8^x is decreasing, so it is below 0.3 to the right of that point.

Example 7. ${}^{1/2}\log(x-1) \geq -2$. Domain: $x > 1$. Equation: ${}^{1/2}\log(x-1) = -2 \iff x - 1 = \left(\frac{1}{2}\right)^{-2} = 4 \iff x = 5$. The function ${}^{1/2}\log$ is decreasing, so the inequality holds to the *left* of $x = 5$, within the domain: $1 < x \leq 5$.

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8. Half-life and doubling time

If $N(t) = b \cdot g^t$ with $g > 1$, the **doubling time** T_2 satisfies $g^{T_2} = 2$, so

$$T_2 = {}^g\log 2 = \frac{\ln 2}{\ln g}.$$

It does not depend on b or on the starting moment: from any moment, the amount doubles in the same time. That is a defining feature of exponential growth.

For decay ($0 < g < 1$), the **half-life** (*halveringstijd*) $T_{1/2}$ satisfies $g^{T_{1/2}} = \frac{1}{2}$, so $T_{1/2} = \dfrac{\ln \frac{1}{2}}{\ln g} = \dfrac{-\ln 2}{\ln g}$.

Conversely, if the half-life is T, the growth factor per time unit is $g = \left(\frac{1}{2}\right)^{1/T}$, and

$$N(t) = N_0 \cdot \left(\tfrac{1}{2}\right)^{t/T}.$$

Example 8. Iodine-131 has a half-life of 8.0 days. Growth factor per day: $0.5^{1/8} \approx 0.917$. Time until only 10% is left: $\left(\frac{1}{2}\right)^{t/8} = 0.1 \Rightarrow t = 8 \cdot \dfrac{\ln 0.1}{\ln 0.5} \approx 26.6$ days.

> **Going further: the rule of 70.** For growth of $p\%$ per year with small p, the doubling time is about $\dfrac{70}{p}$ years. Reason: $T_2 = \dfrac{\ln 2}{\ln(1 + p/100)}$, $\ln 2 \approx 0.693$, and for small h, $\ln(1+h) \approx h$ (the graph of ln at $x = 1$ has slope 1; you will prove this in stage D). So $T_2 \approx \dfrac{0.693}{p/100} \approx \dfrac{70}{p}$. Check: at 3.5% the exact value is about 20.1 years; the rule gives 20.

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9. Logarithmic scales

When quantities range over many powers of ten, we plot or report their logarithm.

Decibels. Sound level $L = 10 \cdot \log \dfrac{I}{I_0}$ dB, with $I_0 = 10^{-12}$ W/m² the hearing threshold. Doubling the intensity adds $10 \log 2 \approx 3$ dB. Ten times the intensity adds exactly 10 dB.

pH. $\text{pH} = -\log[\text{H}^+]$. A drop of 1 in pH means ten times as many hydrogen ions.

Semi-log graphs (*enkellogaritmisch papier*). If $N = b \cdot g^t$, then

$$\log N = \log b + t \log g,$$

which is *linear* in t. So on a graph with a logarithmic vertical axis, exponential growth is a straight line, with intercept $\log b$ and slope $\log g$. This is how scientists test whether data grow exponentially.

Example 9. On a semi-log graph the line through the points $(0,\ 2)$ and $(4,\ 3.2)$ is drawn, where the vertical coordinate is $\log N$. Then $\log N = 2 + 0.3t$, so $N = 10^2 \cdot (10^{0.3})^t \approx 100 \cdot 1.995^t$. The quantity roughly doubles every unit of time (because $10^{0.3} \approx 2$).

> **Going further: log-log graphs.** If both axes are logarithmic, a power function $N = c \cdot t^p$ becomes a straight line: $\log N = \log c + p \log t$, with slope p. Kepler's third law ($T^2 \propto a^3$) shows up as a line with slope $\frac{3}{2}$.

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10. Comparing growth models

A frequent CE question asks *when one quantity overtakes another*.

Example 10. Town A has 500 inhabitants and grows by 8% per year; town B has 2000 inhabitants and grows by 2% per year. When does A overtake B?

$$500 \cdot 1.08^t = 2000 \cdot 1.02^t \iff \left(\frac{1.08}{1.02}\right)^t = 4 \iff t = \frac{\ln 4}{\ln \frac{1.08}{1.02}} \approx 24.3.$$

Dividing both sides by $500 \cdot 1.02^t$ collected all the t's into one power. Since $\frac{1.08}{1.02} > 1$, the left side is increasing, so A is larger for all $t > 24.3$: after about 24 years and 3 months.

This also shows a general fact: **every exponential growth eventually beats every other exponential growth with a smaller factor, whatever the starting values**. And every exponential growth eventually beats every power of t: 1.01^t overtakes t^{100}, though only after a very long time. (Problem 23 is a small example.)

Exponential versus logarithmic graphs

The graphs of $y = g^x$ and $y = {}^g\log x$ are mirror images in $y = x$. For $g = 2$ or $g = 10$ they do not intersect: g^x lies above $y = x$ everywhere, and ${}^g\log x$ below. But for g close to 1 they do: $y = (\sqrt{2})^x$ and $y = {}^{\sqrt{2}}\log x$ meet at $(2, 2)$ and $(4, 4)$, because $(\sqrt{2})^2 = 2$ and $(\sqrt{2})^4 = 4$. The borderline case is $g = e^{1/e} \approx 1.444$, where the two graphs just touch the line $y = x$ at $x = e$. You will be able to prove that with the derivatives of stage D.

> **Going further: what does $2^{\sqrt{2}}$ mean?** For rational exponents, $2^{p/q} = \sqrt[q]{2^p}$. For an irrational exponent like $\sqrt{2} = 1.41421\ldots$, take the rational approximations $2^{1.4}, 2^{1.41}, 2^{1.414}, \ldots$. They form an increasing sequence that is bounded (by $2^{1.5}$), and a theorem from real analysis (the completeness of the real numbers) says such a sequence has a limit. That limit is $2^{\sqrt{2}} \approx 2.665$. All the power rules carry over by taking limits. This is why the graph of 2^x has no gaps.

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11. Common mistakes, collected

Wrong	Right	Why
$\log(a+b) = \log a + \log b$	no rule for $\log(a+b)$	the product rule is about $\log(ab)$
$\dfrac{\log a}{\log b} = \log \dfrac{a}{b}$	$\dfrac{\log a}{\log b} = {}^{b}\log a$	a quotient of logs is a change of base
$(\log x)^2 = 2\log x$	$\log(x^2) = 2\log x$	the power rule needs the power *inside*
$2 \cdot 3^x = 6^x$	$2 \cdot 3^x$ stays as it is	only equal bases can be combined
$0.8^x < 0.3 \Rightarrow x < {}^{0.8}\log 0.3$	$x > {}^{0.8}\log 0.3$	base below 1: decreasing function
${}^{3}\log x + {}^{3}\log(x-8) = 2$ has solutions 9 and -1	only 9	domain of the original equation
growth 4% per year $\Rightarrow$ 40% per decade	$1.04^{10} \approx 1.48$: 48%	factors multiply, they do not add

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12. A complete CE-style example

Example 11. Given $f(x) = \ln(x^2 - 4)$.

a. Give the domain. $x^2 - 4 > 0 \iff x < -2$ or $x > 2$.

b. Solve exactly: $f(x) = \ln 5$. ln is one-to-one, so $x^2 - 4 = 5$, $x = \pm 3$. Both lie in the domain.

c. Solve exactly: $f(x) < \ln 12$. ln is increasing, so $x^2 - 4 < 12$ together with $x^2 - 4 > 0$: $4 < x^2 < 16$, i.e. $2 < |x| < 4$. **Answer:** $-4 < x < -2$ or $2 < x < 4$.

d. The graph of f is symmetric in the y-axis. Toon aan. $f(-x) = \ln((-x)^2 - 4) = \ln(x^2 - 4) = f(x)$, and the domain is symmetric, so f is even.

Summary

- ${}^{g}\log x$ is the exponent that turns g into x; it is the inverse of g^x.
- Log rules = power rules read backwards. Learn the proofs.
- Change of base: ${}^{g}\log a = \dfrac{\log a}{\log g}$.
- e = limit of $(1 + \frac{1}{n})^n$; $bg^t = be^{t \ln g}$.
- Equations: equal bases, take logs, use the definition, or substitute $u = g^x$. Check the domain.
- Inequalities: a base below 1 reverses the inequality.
- Doubling time ${}^{g}\log 2$; half-life ${}^{g}\log \frac{1}{2}$.
- On a semi-log graph, exponential growth is a straight line.